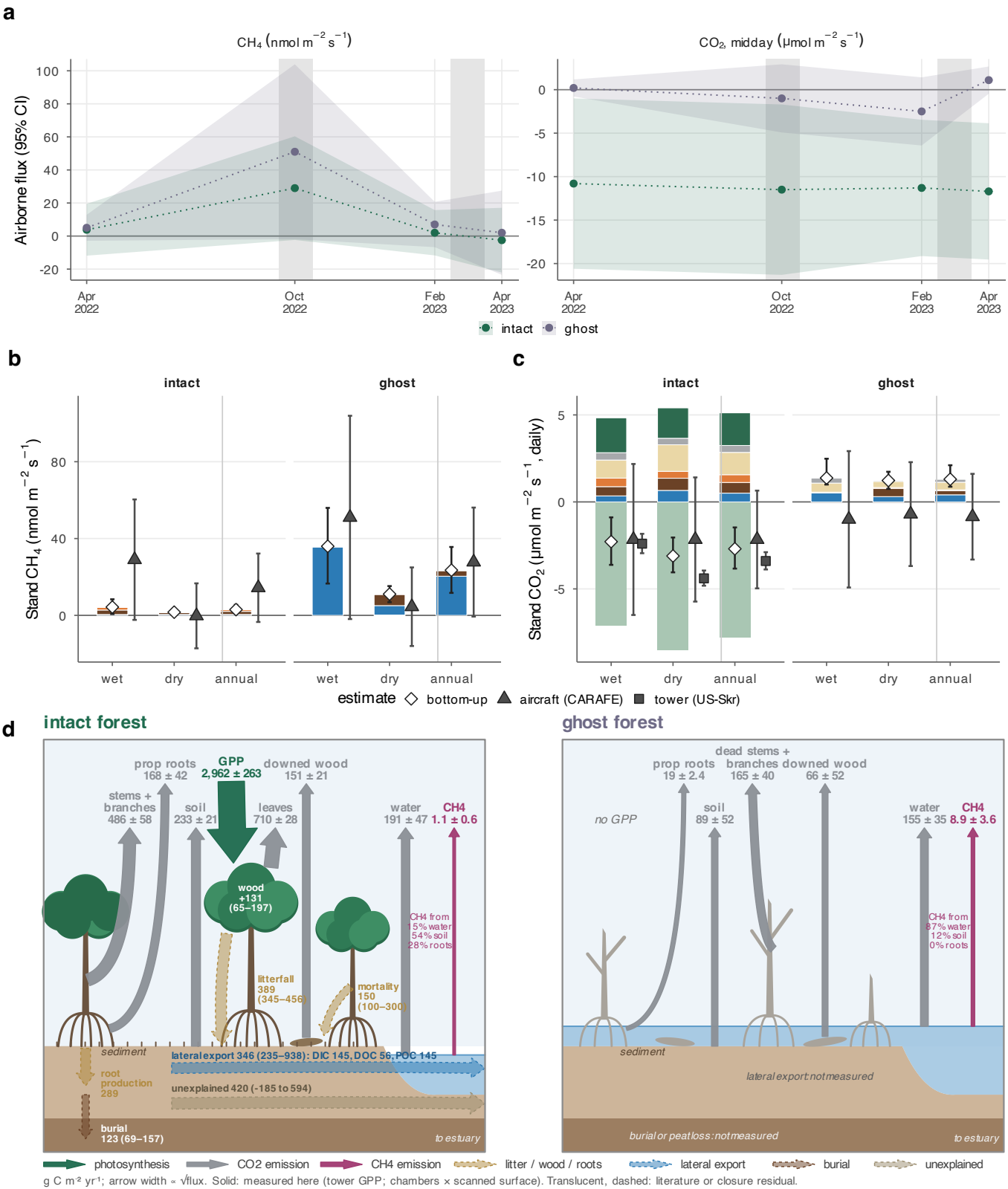

a
CH4 (nmol m−2 s−1)
CO2, midday (µmol m−2 s−1)
Airborne flux (95% CI)
Apr 2022
Oct 2022
Feb 2023
Apr 2023
intact
ghost
b
intact
ghost
Stand CH4 (nmol m−2 s−1)
wet
dry
annual
c
Stand CO2 (µmol m−2 s−1, daily)
estimate
bottom-up
aircraft (CARAFE)
tower (US-Skr)
d
intact forest
prop roots 168 ± 42
GPP 2,962 ± 263
downed wood 151 ± 21
stems + branches 486 ± 58
soil 233 ± 21
leaves 710 ± 28
water 191 ± 47
CH4 1.1 ± 0.6
wood +131 (65–197)
litterfall 389 (345–456)
mortality 150 (100–300)
CH4 from 15% water 54% soil 28% roots
sediment
root production 289
lateral export 346 (235–938): DIC 145, DOC 56, POC 145
unexplained 420 (-185 to 594)
burial 123 (69–157)
to estuary
ghost forest
dead stems + prop roots 19 ± 2.4
branches 165 ± 40
downed wood 66 ± 52
no GPP
soil 89 ± 52
water 155 ± 35
CH4 8.9 ± 3.6
CH4 from 87% water 12% soil 0% roots
sediment
lateral export: not measured
burial or peat loss: not measured
to estuary
photosynthesis
CO2 emission
CH4 emission
litter / wood / roots
lateral export
burial
unexplained
g C m-2 yr-1; arrow width ∝ √flux. Solid: measured here (tower GPP; chambers × scanned surface). Translucent, dashed: literature or closure residual.